Properties of Purine bases

1. Purines are basic in nature. They are sparingly soluble in water.
2. They also absorb UV light at 260 nm. This property is used to detect and estimate DNA and RNA in biological solutions.
3. They are capable of forming hydrogen bonds with pyrimidine nucleotides in nucleic acids.
4. They exhibit keto –enol tautomerism.

7.2.2 Sugars

There are two types of sugars present in nucleic acids. They are ribose and deoxyribose (Fig.7.3). Based on the sugar moiety present in the nucleic acids, they are classified as Deoxy ribonucleic acid (DNA) and Ribonucleic acid (RNA). DNA is present in the nucleus, mitochondria and chloroplasts. RNA is present in the nucleus, nucleolus, ribosome and cytoplasm. Ribose and deoxy ribose have differences in their properties. Deoxy ribose is less reactive in nature, when compared to ribose.

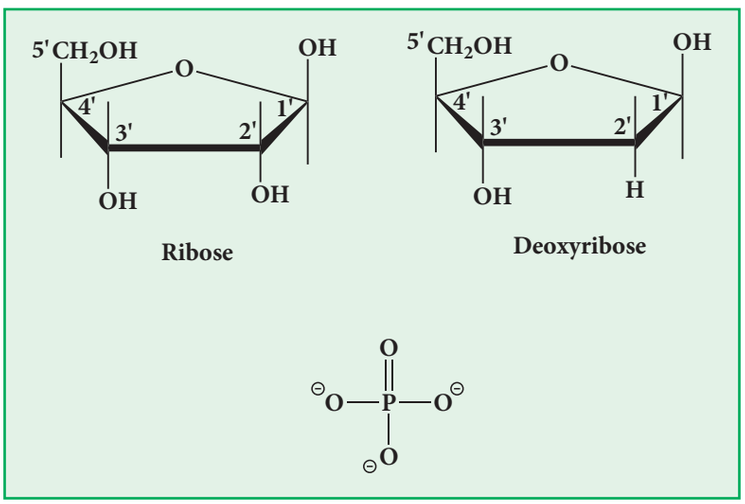

Figure 7.3 Structure of ribose, deoxy ribose and phosphate

7.2.3 Phosphate

Phosphoric acid forms phospho-diester linkage between nucleosides. Based on the number of phosphate group present in the nucleotide, they are classified as monophosphates, diphosphates and triphosphates.

unit 7.indd 179 21-05-2018 15:11:14